

ECO 7424: Econometric Models and Methods

University of Florida - Spring 2025

Instructor

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Office hours: Wednesdays, 1:40-3:40pm

Course description

ECO 7424 is the second course in the 1st-year PhD econometrics sequence. It builds on ECO 7415 and introduces some of the foundational models and methods in econometrics. The first half of the course will focus on linear regression, while the second half will consider topics including clustering, the bootstrap, endogeneity, and the generalized method of moments.

Prerequisites

The prerequisite for the course is ECO 7415: Statistical Methods in Economics.

Class meetings and office hours

Classes are scheduled to take place from 11:45am-1:40pm on Mondays and Wednesdays. They will occur in Matherly Hall Room 14.

I will hold office hours on Wednesdays from 1:40-3:40pm. My office hours will occur either in person at my office or over Zoom. The Zoom link can be found on the course webpage on Canvas. If you would like to schedule a meeting at a different time, please email me.

Course modality

This is a fully in-person course. All midterm exams will take place in person, during our scheduled class times.

I will record the class sessions via Zoom. If you are unable to come to class because you are feeling sick or because you have another valid excuse, you may contact me for the Zoom link. You will then be able to participate remotely.

At the end of each week, I will post the Zoom recordings on Canvas, together with lecture slides and lecture notes. You may consult these as you wish.

Textbook

The course will closely follow the graduate econometrics text, *Econometrics*, from Bruce Hansen. This text is available online at www.ssc.wisc.edu/~bhansen/econometrics. There are a number of other books that cover similar material, including:

- Cameron and Trivedi (2005), *Microeconometrics*
- Davidson and MacKinnon (2004), *Econometric Theory and Methods*
- Greene (2001), *Econometric Analysis*
- Hayashi (2001), *Econometrics*
- Wooldridge (2010), *Econometric Analysis of Cross Section and Panel Data*

When you find a concept confusing, it is often useful to consult these other texts for an alternative explanation.

Midterm exams

The course will have two non-cumulative midterm exams. During the class session before each exam, I will hold a review session. The dates and topics for the exams are shown below:

Table 1: Exam details

Exam	Date	Topics
1	03/12	Linear regression
2	04/23	Additional topics

Problem sets

The course will have frequent problem sets. Late submissions will not be accepted. However, we will drop the assignment with the lowest grade.

Problem sets should be typed and submitted via Canvas. I recommend doing so using LaTeX. An easy way to introduce yourself to LaTeX is through a program called LyX. LyX is free and open source, and it allows producing documents in LaTeX without having to know all the code. LyX is available at: <https://www.lyx.org/Download>.

Students are allowed to work together on problem sets. However, each student must write up his/her own submission.

Participation

I would like students to actively participate in class. Our class meetings are two hours long. The best way to stay engaged during such a lengthy period is by being an active participant. Students should feel free to ask questions whenever something is unclear. In addition, I will often ask questions to the class to make sure that everyone is following.

Grades

The grading structure for the course will be the following:

Table 2: Grading details

Component	Percent
Problem sets	15%
Exam 1	40%
Exam 2	45%

Grading will be consistent with [UF policies regarding grade determination](#).

The final letter grade distribution will be determined as follows:

Table 3: Grade scale

Score	Grade
93-100	A
90-92	A-
87-89	B+
83-86	B
80-82	B-
77-79	C+
73-76	C
70-72	C-
67-69	D+
60-66	D
0-59	E

Professionalism and Honor Code

Students are bound to not cheat or plagiarize, and are bound by The Honor Pledge which states, “We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honor and integrity by abiding by the Honor Code. On all work submitted for credit by students at the University of Florida, the following pledge is either required or implied: ‘On my honor, I have neither given nor received unauthorized aid in doing this assignment.’”

You should familiarize yourself with the [UF Student Honor Code](#). Cheating and plagiarism are not the only violations of this policy. Importantly, ignorance of a policy is not a valid reason for violating it.

Students requiring accommodations

Students with disabilities requesting accommodations should first register with the [Disabilities Resource Center](#) (352-392-8565), providing appropriate documentation. Once registered, students

will receive an accommodation letter that can be presented to the instructor when requesting accommodations. Please register at the beginning of the course if seeking accommodations.

Course evaluations

Students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online via GatorEvals. Students will be notified when the evaluation period opens, and can complete evaluations through the email they receive from GatorEvals, in their Canvas course menu under GatorEvals, or via ufl.bluer.com/ufl.

UF Teaching Center

The UF Teaching Center offers guidance on study skills and tutoring services. You can find more information at: <https://umatter.ufl.edu/office/teaching-center>.

In-class recording

Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled. The only allowable purposes are (1) for personal educational use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil proceeding. All other purposes are prohibited. Specifically, students may not publish recorded lectures without the written consent of the instructor. A "class lecture" is an educational presentation intended to inform or teach enrolled students about a particular subject, including any instructor-led discussions that form part of the presentation, and delivered by any instructor hired or appointed by the University, or by a guest instructor, as part of a University of Florida course. A class lecture does not include lab sessions, student presentations, clinical presentations such as patient history, academic exercises involving solely student participation, assessments (quizzes, tests, exams), field trips, private conversations between students in the class or between a student and the faculty or lecturer during a class session.

Make-up assignments

Make-up assignments and exams will be arranged only for absences that are explicitly covered by the [UF Attendance Policy](#). Whenever possible, you should reach out at least five business days in advance to arrange a make-up assignment or exam. Of course, this will not always be possible. Unforeseen absences and emergencies occur and can be excused without such advance notice. In most cases, you will be asked to provide evidence or documentation of an absence that is explicitly excused by the UF Attendance Policy. Absences related to religious holidays and worship do not require this documentation.

Course structure

A tentative list of the topics that we will cover is presented below.

Linear regression

1. Conditional expectations and projection
 - Hansen Chapter 2.1-2.25
2. The algebra of least squares
 - Hansen Chapter 3
3. Least squares regression
 - Hansen Chapter 4.1-4.22
4. Asymptotic theory for least squares
 - Hansen Chapter 6, 7

Additional topics

1. Clustering
 - Hansen Chapter 4.23-4.25
2. The bootstrap
 - Hansen Chapter 10
3. Endogeneity and IV
 - Hansen Chapter 12.1-12.12
4. Generalized method of moments
 - Hansen Chapter 13